Similar questions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The matrix $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}$ represents a transformation T.

(a) The line $y = kx$ is invariant under T. Determine the value of k. **[4]**

(b) Determine whether $y = kx$ is a line of invariant points. **[1]**

(c) Find the image of the point $(2, -2)$ under T. **[1]**

2

(a) $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$. Describe fully the transformation represented by $\mathbf{A}$, and show that the area of any shape is unchanged by it. **[3]**

$\mathbf{B} = \begin{pmatrix} 7 & 3 \\ 10 & 5 \end{pmatrix}$ is the result of applying three transformations in this order: a shear that leaves the y-axis invariant and maps $(1,1)$ to $(1,3)$; then the transformation represented by $\mathbf{A}$; then a stretch of scale factor p that leaves the x-axis invariant.

(b) Determine p. **[4]**

Question 1

(a) $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 4+k \\ -1+2k \end{pmatrix}$. Invariant needs $-1+2k = k(4+k)$, so $k^2+2k+1=0$ and $k=-1$

(b) $T(1,-1) = (3,-3) \neq (1,-1)$, so the points are not fixed: $y=-x$ is not a line of invariant points

(c) $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$

Question 2

(a) A shear with the x-axis invariant (a line of invariant points), mapping (x,y) to $(x+3y,y)$, e.g. $(0,1) \mapsto (3,1)$. $\det \mathbf{A} = 1$ so areas are unchanged

(b) The first shear is $\begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$ with $(1,1) \mapsto (1,1+s) = (1,3)$, so $s=2$. The stretch is $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. The combined matrix is $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ 2p & p \end{pmatrix}$. Comparing with $\mathbf{B}$: $p=5$